

Solar Portfolio Assessment

10 candidate sites

1	Solar Star Corridor 3	90	High	HIGH
2	Edwards-Sanborn Corridor 5	87	High	HIGH
3	Solar Star Corridor 1	86	High	HIGH
4	San Joaquin Valley Ag 2 (Buttonwillow)	86	High	HIGH
5	San Joaquin Valley Ag 1 (W of Wasco)	84	High	HIGH
6	Solar Star Corridor 2	79	High	HIGH
7	Edwards-Sanborn Corridor 4	78	High	HIGH
8	San Joaquin Valley Ag 3 (Semitropic)	76	High	HIGH
9	San Joaquin Valley Ag 4 (near Shafter)	Excluded	Excluded	HIGH
10	San Joaquin Valley Ag 5 (near Lost Hills)	Excluded	Excluded	HIGH

Solar Star Corridor 3

Solar Site Suitability Assessment

LOCATION: Solar Star Corridor 3 · NERC REGION: WECC

SITE SCORE

90 / 100

High

CONFIDENCE

95 / 100

HIGH

"This assessment is based on authoritative data for all major criteria."

DATA GAPS

- Environmental Justice screening data is unavailable at this location. The EPA EJScreen tool has been discontinued.

INTERCONNECTION CONTEXT

Nearest substation: Polaris Substation (0.9 mi)

Existing capacity (within 50 km): 4375.9 MW across 141 operating plants (EIA Form 860).

Queue activity (n/a): 0 active solar interconnection requests totaling 0 MW within 50 km.

NOTE: Informational context from public data, not an interconnection study. Actual capacity availability requires an interconnection application with the ISO/RTO. Queue data: LBNL "Queued Up" 2025, county-centroid precision.

EXCLUSION SCREENING

Critical habitat	✓ pass	Critical Habitat Designations
Protected areas	✓ pass	Protected Areas Database (PAD-US)
Steep slope	✓ pass	USGS 3D Elevation Program
Developed land	✓ pass	National Land Cover Database 2021
Wetlands	✓ pass	National Wetlands Inventory

SCORED CRITERIA

Transmission proximity	0.42	98	HIFLD Transmission Lines	HIGH
Solar resource (GHI)	0.10	93	NREL PVWatts v8	HIGH
Terrain slope	0.17	95	USGS 3D Elevation Program	HIGH
Road access	0.10	50	OpenStreetMap Roads	HIGH
Terrain aspect	0.08	100	USGS 3D Elevation Program	HIGH

Land cover type	0.08	60	National Land Cover Database 2021	HIGH
Soil buildability	0.05	100	USDA SSURGO Soil Data	HIGH
Flood zone	0.05	50	National Flood Hazard Layer	HIGH
Environmental Justice	0.00	—	—	NONE

WEIGHT PROFILE: WECC — calibrated against 100 EIA Form 860 installations

Edwards-Sanborn Corridor 5

Solar Site Suitability Assessment

LOCATION: Edwards-Sanborn Corridor 5 · NERC REGION: WECC

SITE SCORE

87 / 100

High

CONFIDENCE

95 / 100

HIGH

"This assessment is based on authoritative data for all major criteria."

DATA GAPS

- Environmental Justice screening data is unavailable at this location. The EPA EJScreen tool has been discontinued.

INTERCONNECTION CONTEXT

Nearest substation: Polaris Substation (0.8 mi)

Existing capacity (within 50 km): 4385.4 MW across 142 operating plants (EIA Form 860).

Queue activity (n/a): 0 active solar interconnection requests totaling 0 MW within 50 km.

NOTE: Informational context from public data, not an interconnection study. Actual capacity availability requires an interconnection application with the ISO/RTO. Queue data: LBNL "Queued Up" 2025, county-centroid precision.

EXCLUSION SCREENING

Critical habitat	✓ pass	Critical Habitat Designations
Protected areas	✓ pass	Protected Areas Database (PAD-US)
Steep slope	✓ pass	USGS 3D Elevation Program
Developed land	✓ pass	National Land Cover Database 2021
Wetlands	✓ pass	National Wetlands Inventory

SCORED CRITERIA

Transmission proximity	0.42	96	HIFLD Transmission Lines	HIGH
Solar resource (GHI)	0.10	93	NREL PVWatts v8	HIGH
Terrain slope	0.17	97	USGS 3D Elevation Program	HIGH
Road access	0.10	50	OpenStreetMap Roads	HIGH
Terrain aspect	0.08	100	USGS 3D Elevation Program	HIGH

Land cover type	0.08	60	National Land Cover Database 2021	HIGH
Soil buildability	0.05	60	USDA SSURGO Soil Data	HIGH
Flood zone	0.05	100	National Flood Hazard Layer	HIGH
Environmental Justice	0.00	—	—	NONE

WEIGHT PROFILE: WECC — calibrated against 100 EIA Form 860 installations

Solar Star Corridor 1

Solar Site Suitability Assessment

LOCATION: Solar Star Corridor 1 · NERC REGION: WECC

SITE SCORE

86 / 100

High

CONFIDENCE

95 / 100

HIGH

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DATA GAPS

- Environmental Justice screening data is unavailable at this location. The EPA EJScreen tool has been discontinued.

INTERCONNECTION CONTEXT

Nearest substation: (unnamed substation) (2.1 mi, 220 kV)

Existing capacity (within 50 km): 4375.9 MW across 141 operating plants (EIA Form 860).

Queue activity (n/a): 0 active solar interconnection requests totaling 0 MW within 50 km.

NOTE: Informational context from public data, not an interconnection study. Actual capacity availability requires an interconnection application with the ISO/RTO. Queue data: LBNL "Queued Up" 2025, county-centroid precision.

EXCLUSION SCREENING

Critical habitat	✓ pass	Critical Habitat Designations
Protected areas	✓ pass	Protected Areas Database (PAD-US)
Steep slope	✓ pass	USGS 3D Elevation Program
Developed land	✓ pass	National Land Cover Database 2021
Wetlands	✓ pass	National Wetlands Inventory

SCORED CRITERIA

Transmission proximity	0.42	91	HIFLD Transmission Lines	HIGH
Solar resource (GHI)	0.10	93	NREL PVWatts v8	HIGH
Terrain slope	0.17	95	USGS 3D Elevation Program	HIGH
Road access	0.10	50	OpenStreetMap Roads	HIGH
Terrain aspect	0.08	100	USGS 3D Elevation Program	HIGH

Land cover type	0.08	60	National Land Cover Database 2021	HIGH
Soil buildability	0.05	100	USDA SSURGO Soil Data	HIGH
Flood zone	0.05	50	National Flood Hazard Layer	HIGH
Environmental Justice	0.00	—	—	NONE

WEIGHT PROFILE: WECC — calibrated against 100 EIA Form 860 installations

San Joaquin Valley Ag 2 (Buttonwillow)

Solar Site Suitability Assessment

LOCATION: San Joaquin Valley Ag 2 (Buttonwillow) · NERC REGION: WECC

SITE SCORE

86 / 100

High

CONFIDENCE

95 / 100

HIGH

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DATA GAPS

- Environmental Justice screening data is unavailable at this location. The EPA EJScreen tool has been discontinued.

INTERCONNECTION CONTEXT

Nearest substation: (unnamed substation) (4.9 mi)

Existing capacity (within 50 km): 1245.8 MW across 53 operating plants (EIA Form 860).

Queue activity (n/a): 0 active solar interconnection requests totaling 0 MW within 50 km.

NOTE: Informational context from public data, not an interconnection study. Actual capacity availability requires an interconnection application with the ISO/RTO. Queue data: LBNL "Queued Up" 2025, county-centroid precision.

EXCLUSION SCREENING

Critical habitat	✓ pass	Critical Habitat Designations
Protected areas	✓ pass	Protected Areas Database (PAD-US)
Steep slope	✓ pass	USGS 3D Elevation Program
Developed land	✓ pass	National Land Cover Database 2021
Wetlands	✓ pass	National Wetlands Inventory

SCORED CRITERIA

Transmission proximity	0.42	97	HIFLD Transmission Lines	HIGH
Solar resource (GHI)	0.10	65	NREL PVWatts v8	HIGH
Terrain slope	0.17	96	USGS 3D Elevation Program	HIGH
Road access	0.10	50	OpenStreetMap Roads	HIGH
Terrain aspect	0.08	100	USGS 3D Elevation Program	HIGH

Land cover type	0.08	60	National Land Cover Database 2021	HIGH
Soil buildability	0.05	100	USDA SSURGO Soil Data	HIGH
Flood zone	0.05	100	National Flood Hazard Layer	HIGH
Environmental Justice	0.00	—	—	NONE

WEIGHT PROFILE: WECC — calibrated against 100 EIA Form 860 installations

San Joaquin Valley Ag 1 (W of Wasco)

Solar Site Suitability Assessment

LOCATION: San Joaquin Valley Ag 1 (W of Wasco) · NERC REGION: WECC

SITE SCORE

84 / 100

High

CONFIDENCE

95 / 100

HIGH

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DATA GAPS

- Environmental Justice screening data is unavailable at this location. The EPA EJScreen tool has been discontinued.

INTERCONNECTION CONTEXT

Nearest substation: (unnamed substation) (1.7 mi)

Existing capacity (within 50 km): 664.9 MW across 58 operating plants (EIA Form 860).

Queue activity (n/a): 0 active solar interconnection requests totaling 0 MW within 50 km.

NOTE: Informational context from public data, not an interconnection study. Actual capacity availability requires an interconnection application with the ISO/RTO. Queue data: LBNL "Queued Up" 2025, county-centroid precision.

EXCLUSION SCREENING

Critical habitat	✓ pass	Critical Habitat Designations
Protected areas	✓ pass	Protected Areas Database (PAD-US)
Steep slope	✓ pass	USGS 3D Elevation Program
Developed land	✓ pass	National Land Cover Database 2021
Wetlands	✓ pass	National Wetlands Inventory

SCORED CRITERIA

Transmission proximity	0.42	85	HIFLD Transmission Lines	HIGH
Solar resource (GHI)	0.10	65	NREL PVWatts v8	HIGH
Terrain slope	0.17	98	USGS 3D Elevation Program	HIGH
Road access	0.10	50	OpenStreetMap Roads	HIGH
Terrain aspect	0.08	100	USGS 3D Elevation Program	HIGH

Land cover type	0.08	90	National Land Cover Database 2021	HIGH
Soil buildability	0.05	100	USDA SSURGO Soil Data	HIGH
Flood zone	0.05	100	National Flood Hazard Layer	HIGH
Environmental Justice	0.00	—	—	NONE

WEIGHT PROFILE: WECC — calibrated against 100 EIA Form 860 installations

Solar Star Corridor 2

Solar Site Suitability Assessment

LOCATION: Solar Star Corridor 2 · NERC REGION: WECC

SITE SCORE

79 / 100

High

CONFIDENCE

95 / 100

HIGH

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DATA GAPS

- Environmental Justice screening data is unavailable at this location. The EPA EJScreen tool has been discontinued.

INTERCONNECTION CONTEXT

Nearest substation: (unnamed substation) (1.0 mi, 220 kV)

Existing capacity (within 50 km): 4375.9 MW across 141 operating plants (EIA Form 860).

Queue activity (n/a): 0 active solar interconnection requests totaling 0 MW within 50 km.

NOTE: Informational context from public data, not an interconnection study. Actual capacity availability requires an interconnection application with the ISO/RTO. Queue data: LBNL "Queued Up" 2025, county-centroid precision.

EXCLUSION SCREENING

Critical habitat	✓ pass	Critical Habitat Designations
Protected areas	✓ pass	Protected Areas Database (PAD-US)
Steep slope	✓ pass	USGS 3D Elevation Program
Developed land	✓ pass	National Land Cover Database 2021
Wetlands	✓ pass	National Wetlands Inventory

SCORED CRITERIA

Transmission proximity	0.42	82	HIFLD Transmission Lines	HIGH
Solar resource (GHI)	0.10	92	NREL PVWatts v8	HIGH
Terrain slope	0.17	94	USGS 3D Elevation Program	HIGH
Road access	0.10	50	OpenStreetMap Roads	HIGH
Terrain aspect	0.08	100	USGS 3D Elevation Program	HIGH

Land cover type	0.08	60	National Land Cover Database 2021	HIGH
Soil buildability	0.05	100	USDA SSURGO Soil Data	HIGH
Flood zone	0.05	35	National Flood Hazard Layer	HIGH
Environmental Justice	0.00	—	—	NONE

WEIGHT PROFILE: WECC — calibrated against 100 EIA Form 860 installations

Edwards-Sanborn Corridor 4

Solar Site Suitability Assessment

LOCATION: Edwards-Sanborn Corridor 4 · NERC REGION: WECC

SITE SCORE

78 / 100

High

CONFIDENCE

95 / 100

HIGH

"This assessment is based on authoritative data for all major criteria."

DATA GAPS

- Environmental Justice screening data is unavailable at this location. The EPA EJScreen tool has been discontinued.

INTERCONNECTION CONTEXT

Nearest substation: (unnamed substation) (0.6 mi, 220 kV)

Existing capacity (within 50 km): 4411.9 MW across 137 operating plants (EIA Form 860).

Queue activity (n/a): 0 active solar interconnection requests totaling 0 MW within 50 km.

NOTE: Informational context from public data, not an interconnection study. Actual capacity availability requires an interconnection application with the ISO/RTO. Queue data: LBNL "Queued Up" 2025, county-centroid precision.

EXCLUSION SCREENING

Critical habitat	✓ pass	Critical Habitat Designations
Protected areas	✓ pass	Protected Areas Database (PAD-US)
Steep slope	✓ pass	USGS 3D Elevation Program
Developed land	✓ pass	National Land Cover Database 2021
Wetlands	✓ pass	National Wetlands Inventory

SCORED CRITERIA

Transmission proximity	0.42	86	HIFLD Transmission Lines	HIGH
Solar resource (GHI)	0.10	90	NREL PVWatts v8	HIGH
Terrain slope	0.17	92	USGS 3D Elevation Program	HIGH
Road access	0.10	50	OpenStreetMap Roads	HIGH
Terrain aspect	0.08	70	USGS 3D Elevation Program	HIGH

Land cover type	0.08	60	National Land Cover Database 2021	HIGH
Soil buildability	0.05	100	USDA SSURGO Soil Data	HIGH
Flood zone	0.05	35	National Flood Hazard Layer	HIGH
Environmental Justice	0.00	—	—	NONE

WEIGHT PROFILE: WECC — calibrated against 100 EIA Form 860 installations

San Joaquin Valley Ag 3 (Semitropic)

Solar Site Suitability Assessment

LOCATION: San Joaquin Valley Ag 3 (Semitropic) · NERC REGION: WECC

SITE SCORE

76 / 100

High

CONFIDENCE

95 / 100

HIGH

"This assessment is based on authoritative data for all major criteria."

DATA GAPS

- Environmental Justice screening data is unavailable at this location. The EPA EJScreen tool has been discontinued.

INTERCONNECTION CONTEXT

Nearest substation: (unnamed substation) (4.8 mi)

Existing capacity (within 50 km): 1258.1 MW across 47 operating plants (EIA Form 860).

Queue activity (n/a): 0 active solar interconnection requests totaling 0 MW within 50 km.

NOTE: Informational context from public data, not an interconnection study. Actual capacity availability requires an interconnection application with the ISO/RTO. Queue data: LBNL "Queued Up" 2025, county-centroid precision.

EXCLUSION SCREENING

Critical habitat	✓ pass	Critical Habitat Designations
Protected areas	✓ pass	Protected Areas Database (PAD-US)
Steep slope	✓ pass	USGS 3D Elevation Program
Developed land	✓ pass	National Land Cover Database 2021
Wetlands	✓ pass	National Wetlands Inventory

SCORED CRITERIA

Transmission proximity	0.42	68	HIFLD Transmission Lines	HIGH
Solar resource (GHI)	0.10	65	NREL PVWatts v8	HIGH
Terrain slope	0.17	95	USGS 3D Elevation Program	HIGH
Road access	0.10	50	OpenStreetMap Roads	HIGH
Terrain aspect	0.08	100	USGS 3D Elevation Program	HIGH

Land cover type	0.08	90	National Land Cover Database 2021	HIGH
Soil buildability	0.05	100	USDA SSURGO Soil Data	HIGH
Flood zone	0.05	50	National Flood Hazard Layer	HIGH
Environmental Justice	0.00	—	—	NONE

WEIGHT PROFILE: WECC — calibrated against 100 EIA Form 860 installations

San Joaquin Valley Ag 4 (near Shafter)

Solar Site Suitability Assessment

LOCATION: San Joaquin Valley Ag 4 (near Shafter) · NERC REGION: WECC

SITE SCORE

72 / 100

Excluded

CONFIDENCE

95 / 100

HIGH

"This assessment is based on authoritative data for all major criteria."

DATA GAPS

- Environmental Justice screening data is unavailable at this location. The EPA EJScreen tool has been discontinued.

INTERCONNECTION CONTEXT

Nearest substation: (unnamed substation) (2.5 mi)

Existing capacity (within 50 km): 740.9 MW across 57 operating plants (EIA Form 860).

Queue activity (n/a): 0 active solar interconnection requests totaling 0 MW within 50 km.

NOTE: Informational context from public data, not an interconnection study. Actual capacity availability requires an interconnection application with the ISO/RTO. Queue data: LBNL "Queued Up" 2025, county-centroid precision.

EXCLUSION SCREENING

Critical habitat	✓ pass	Critical Habitat Designations
Protected areas	✓ pass	Protected Areas Database (PAD-US)
Steep slope	✓ pass	USGS 3D Elevation Program
Developed land	✗ EXCLUDED	National Land Cover Database 2021
Wetlands	✓ pass	National Wetlands Inventory

SCORED CRITERIA

Transmission proximity	0.42	74	HIFLD Transmission Lines	HIGH
Solar resource (GHI)	0.10	66	NREL PVWatts v8	HIGH
Terrain slope	0.17	98	USGS 3D Elevation Program	HIGH
Road access	0.10	50	OpenStreetMap Roads	HIGH
Terrain aspect	0.08	100	USGS 3D Elevation Program	HIGH

Land cover type	0.08	0	National Land Cover Database 2021	HIGH
Soil buildability	0.05	100	USDA SSURGO Soil Data	HIGH
Flood zone	0.05	100	National Flood Hazard Layer	HIGH
Environmental Justice	0.00	—	—	NONE

WEIGHT PROFILE: WECC — calibrated against 100 EIA Form 860 installations

San Joaquin Valley Ag 5 (near Lost Hills)

Solar Site Suitability Assessment

LOCATION: San Joaquin Valley Ag 5 (near Lost Hills) · NERC REGION: WECC

SITE SCORE

62 / 100

Excluded

CONFIDENCE

95 / 100

HIGH

"This assessment is based on authoritative data for all major criteria."

DATA GAPS

- Environmental Justice screening data is unavailable at this location. The EPA EJScreen tool has been discontinued.

INTERCONNECTION CONTEXT

Nearest substation: (unnamed substation) (2.2 mi)

Existing capacity (within 50 km): 1248.2 MW across 43 operating plants (EIA Form 860).

Queue activity (n/a): 0 active solar interconnection requests totaling 0 MW within 50 km.

NOTE: Informational context from public data, not an interconnection study. Actual capacity availability requires an interconnection application with the ISO/RTO. Queue data: LBNL "Queued Up" 2025, county-centroid precision.

EXCLUSION SCREENING

Critical habitat	✓ pass	Critical Habitat Designations
Protected areas	✓ pass	Protected Areas Database (PAD-US)
Steep slope	✓ pass	USGS 3D Elevation Program
Developed land	✗ EXCLUDED	National Land Cover Database 2021
Wetlands	✓ pass	National Wetlands Inventory

SCORED CRITERIA

Transmission proximity	0.42	70	HIFLD Transmission Lines	HIGH
Solar resource (GHI)	0.10	63	NREL PVWatts v8	HIGH
Terrain slope	0.17	93	USGS 3D Elevation Program	HIGH
Road access	0.10	50	OpenStreetMap Roads	HIGH
Terrain aspect	0.08	0	USGS 3D Elevation Program	HIGH

Land cover type	0.08	0	National Land Cover Database 2021	HIGH
Soil buildability	0.05	100	USDA SSURGO Soil Data	HIGH
Flood zone	0.05	50	National Flood Hazard Layer	HIGH
Environmental Justice	0.00	—	—	NONE

WEIGHT PROFILE: WECC — calibrated against 100 EIA Form 860 installations